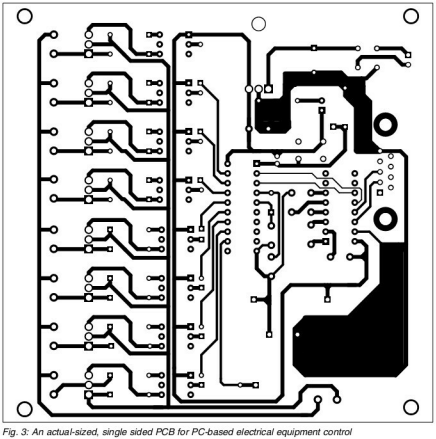


Fig. 3: An actual-sized, single sided PCB for PC-based electrical equipment control

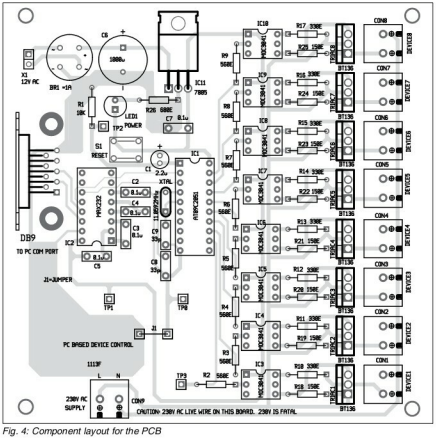


Fig. 4: Component layout for the PCB

plies and other peripheral devices. Now, since it is becoming obsolete, one has to use an external USB-to-RS232 converter to connect to RS232 peripherals.

MOC3041. Opto-TRIAC MOC3041 is a TRIAC driver, with inbuilt zero-crossing detector. The zero-crossing, optically-isolated TRIAC driver is an effective solution to interface the microcontroller ports with the AC power loads. A set of TRIAC driver and TRIAC is used for each load. When a particular load is switched on with the help of the personal computer, the particular TRIAC-driver of the corresponding device/load latches on. This action introduces gate current in the TRIAC (BT136).

BT136. TRIAC BT136 is a solidstate switch used for switching the electrical devices. It is a 3-pin device normally available in a TO220 plastic package. It has three terminals—MT1, MT2 and G (gate).

The TRIAC can be triggered by the current going into the gate. To create a triggering current, a positive or negative voltage has to be applied to the gate with respect to the MT1 terminal. Once triggered, the device continues to conduct until the current drops below a certain threshold called the holding current. When the gate current is discontinued, and if the current flowing between the two main terminals is more than what is called the latching current, the device keeps conducting.

After the TRIAC is turned on, removing the voltage at the gate will not turn the TRIAC to the off state. The only way the TRIAC can be turned off is by reducing the gate current below the holding current or reducing the current to zero by removing the voltage across MT1 and MT2.

When sufficient current of the internal LED (MOC3041) is present, the TRIAC will re-trigger every half cycle of AC line voltage until the internal LED switches off and the power TRIAC has gone through a zero current point. The internal LED is turned on